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Practical approaches for retrofitting plug flow digester and process control to maximize hydrolysis and methane yield from piggery waste

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ABSTRACT

This study demonstrates and quantifies biomethanation improvements from practical schemes of retrofitting anaerobic plug flow (PF) digester for piggery wastewater (1% total solids concentration). First, batch assays were performed to seek the optimal scenario for substrate solubilization. By regulating micro-aeration at different levels, results showed that ORP -420 ± 10 mV promoted maximal hydrolysis efficiency (η_H), to as high as 49.5% above the control (without micro-aeration). However, organics were markedly lost by over oxidation beyond the optimal ORP. Sub-thermophilic (45 °C) was also proven as effective ($p \geq 0.05$) as thermophilic (55 °C) temperature for hydrolysis with η_H 89% higher than that of the ambient (mesophilic) temperature. In continuous digestion studies, the two-stage operations, achieved by adding a pre-hydrolysis (pHY) reactor (HRT 1 day) to the main PF reactor (HRT 4 days), were assessed under four retrofitting schemes. Results showed incremental improvements of each strategy applied and suggested the best overall system configuration of $pHY_{45C}^{-420mV} + PF_{45C}$ with sludge re-hydrolysis. Methane yield was improved by 20.4% over the original PF digester operation. Next Generation Sequencing indicated significant changes in abundance and diversity of the major groups of organisms by the sub-thermophilic temperature that linked to the augmented biomethanation performance. Preliminary cost-benefit analysis is also presented. A full-scale implementation in a pilot farm in Thailand is under development.

1. Introduction

In swine business, manure has been causing serious environmental problems in air, water and soil [1]. It has, nonetheless, been a good and cheap substrate for anaerobic digestion [2] where the biogas produced is utilized as fuel for heat and electricity generation for use in the farm or distributed to the grids [3]. This practice has helped reduce the cost of pork production and to a large degree abate the environmental impacts. By its treatment capability and benefit from energy recovery, anaerobic digesters were constructed in almost all swine farms for quite a long time [4]. This has brought our attention to the existing old anaerobic digesters which are less effective and need an upgrade.

Swine manure itself contains high solids, some of which are slowly biodegradable in anaerobic digester, such as partially digested grain fiber from animal digestive tract [5,6]. The concentration of this piggery wastewater varies from less than 1% to as high as 6% depending on the practice of farm management and water availability [7]. Anaerobic digestion (AD) process consists of four main steps i.e., hydrolysis, acidogenesis, acetogenesis and methanogenesis [3]. Hydrolysis, the first step in anaerobic digestion, is generally a rate-limiting step of AD for the high solids substrate [8]. This slowly biodegradable particulate could accumulate and form a dense layer in the digester's sediment or floating crust causing a loss of its effective volume, which over time lowers its biogas production efficiency. This weakness can be overcome by adding mixing apparatus to the digester to be completely stirred tank reactor (CSTR). Such undertaking unfortunately adds considerable cost to the construction and can only be economically done in newer and larger scale digesters [9]. Other ways to enhance hydrolysis (substrate solubilization) can be done through various pretreatment methods, i.e., physical, chemical, biological means or any combination.

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Physical/chemical pretreatment methods often lead to increased operational costs due to an additional energy for substrate preparation (grinding, cutting, and milling) or chemical additions [10]. On the other hand, biological pretreatment by altering specific conditions or groups of microorganisms through augmentation or stimulation are benign and more economical. These include but not limit to phase-separation, operating temperature (utilizing waste heat), biological oxidation, or microbial inoculation. All of which aim to regulate the reactions and interactions among different groups of microbes in the mixed culture within the digester.

Micro-aeration refers to the introduction of a limited amount of oxygen or air into a reaction vessel with reactants. In case of AD, air could be injected directly to an anaerobic digester as to control the volatile fatty acid (VFA) accumulation. With a very specific amount of air, this approach was reportedly able to stimulate microbial oxidation of VFA and in turn stabilized methanogenic reaction [11,12]. Barati rashvanlou et al. [13] reported positive impacts of limited air supply (0.2 $L_{air}/L_{substrate}$.min) on the digestion of municipal sewage sludge that could promote CH_4 yield by 2.2 times in batch experiment. Certain micro-aerated conditions had been shown to induce higher hydrolytic and acidogenic reactions in high solid fermentation of Napier grass and solid waste [14,15]. However, this strategy in continuous operation must be carefully administered as the methanogens are very sensitive to oxygen and over oxygen supply may result in aerobic oxidation. To regulate microaerobic condition for pretreatment, oxygen or air dosage has always been used. Oxidation-reduction potential (ORP) is in fact a direct representation of the state of reactions and could be a better control parameter. An ORP range under zero is often referred to an intermediate transition between oxidative and reductive environment that reportedly could improve hydrolysis of protein and carbohydrate of various substrates, as well as VFA production [11]. However, there are discrepancies on the proper range of ORP for a purpose of pretreatment since reaction kinetics are dependent on individual substrate characteristics [15,16]. As far as we know, there is rare information on the use of ORP control for pretreatment of wastewaters.

Temperature is an influential parameter for survival and proliferation of different microbes within an anaerobic digester. AD process can take place at different temperatures, normally classified to three ranges; psychrophilic (below 20 °C), mesophilic between 20 and 40 °C (usually 35 °C) and thermophilic between 50 and 65 °C (typically 55 °C) [17]. Although thermophilic condition has certain advantages especially for solid substrate digestion [18], to heat the entire digester to such temperature is too expensive. Since enzymatic activity is highly temperature dependent [19], thermophilic operation give an advantage to the hydrolysis step in AD. Thermophilic digestion relies on energy from waste heat from biogas engines combined with a part of the biogas generated [20]. To retrofit existing unheated digester to 55 °C, large investment on reactor insulation are often too expensive with operational complication. However, our study at a pilot pig farm in Ratchaburi Province, Thailand, using a dual source heat exchanger system demonstrated that waste heats of the biogas engine from cooling water and exhaust gas could afford the temperature of its wastewater holding tank and plug flow digester to a sub-thermophilic range, i.e., 40–45 °C [21]. By not requiring an external heating, this sub-thermophilic temperature range, believed to be less effective for AD [22], could provide a substantial benefit if able to find ways to optimally enhance degradation of this solid-laden piggery wastewater. It, hence, deserves a proper investigation.

With existing anaerobic digesters, retrofitting can be done to upgrade the efficiency and throughput of the system. For single-stage anaerobic digester, all hydrolysis, acidogenesis, acetogenesis and methanogenesis take place in a single unit. When an addition of a front-end reactor is made, establishing a two-stage configuration, AD process can be separated to hydrolysis and acid forming in the first, followed by methane forming in the second stage. This provides superior process efficiency and stability [23,24]. In the first stage, several strategies can be introduced to improve hydrolysis and fermentation through stimulation and augmentation of microbial activity [25]. Utilization of the front unit could be done in several schemes and should be key for system retrofitting and methanation improvement. Proper approaches and optimization are needed for specific substrates.

The aim of this research is to examine the strategies and retrofit schemes to improve biomethanation yield of piggery waste from a standard plug flow digester and to observe how the changes occur from each factor applied. Effects of oxidation-reduction control and temperature to improve hydrolysis-acidogenesis efficiencies were systematically evaluated using a series of batch assays. The optimal conditions identified were subsequently implemented in the continuous digestion study. Four highly practical schemes for retrofitting the plug flow digester were tested to compare CH_4 yield, followed by a preliminary cost-benefit analysis. Microbial community analysis was commenced to observe the changes in individual genus and main groups in AD pathways.

2. Materials and methods

2.1. Piggery wastewater preparation

Fresh swine manure was taken directly from a fattening unit in a local swine farm in Songkhla Province, Thailand, and kept at 4 °C according to Methods for Chemical Analysis of Water and Wastes [26]. It was analyzed and used within 2 days after collection for the batch experiment (Experiment 1, Section 2.3). In order to determine the substrate mass for use, the fresh manure was analyzed for total solids (TS) and volatile solids (VS) which are 253 ± 17 $gTS/kg_{fresh\ wt.}$ and 202 ± 11 $gVS/kg_{fresh\ wt.}$ on wet basis, respectively. Elemental composition of the manure was analyzed as C, H, O, N and S found to be 37.3%, 5.9%, 32.6%, 3.0% and 0.3%, respectively. Its C/N ratio of 15.5 is within a suitable range for anaerobic digestion. Fresh swine manure wastewater (FMW) was prepared by adding deionized water to the fresh manure to make a final TS concentration of 1% or 10,000 mg/L (VS concentration 0.8%) in order to mimic the real piggery wastewater on farm. FMW was used for the batch study as described in Section 2.3 (Experiment 1).

For the long-term continuous digestion study (Experiment 2), a manure wastewater was prepared as feed. The fresh manure from the same farm was spread on a clean concrete floor and sundried, then ground by a crushing machine (FnB Machinery & Solutions, SFL30, Thailand) to 0.2 mm. The manure powder was then dried in an oven at 60 °C until constant weight. The dry manure wastewater (DMW) was prepared to a TS concentration of 1% for use as feed in continuous operation. Characteristics of the FMW and DMW; ORP, pH, NH_4^+, alkalinity, TVFA, fatty acid composition, total chemical oxygen demand (TCOD) and soluble oxygen demand (SCOD), are listed in Table S1 in Supplementary Material. The data in Table S1 are means and standard deviations from the samples prepared for each run during the experiment ($n \geq 5$). Hypothetical methane yield of the DMW from the Biochemical Methane Potential (BMP) assays at 35 and 45 °C were 163 and 177 $mLCH_4/gCOD_{add}$, respectively. Procedure for BMP assay can be found in various works previously published [27,28].

2.2. Inoculum preparation

The inoculum used in the continuous digestion studies (Section 2.4) was the anaerobic sludge taken from a full-scale plug-flow anaerobic digester of a swine farm in Ratchaburi Province, Thailand (N13° 23' 14'', E99° 49' 13'') that was operated at ambient temperature (25–32 °C). The sludge was drawn from the settling zone near the digester outlet. TS and VS were measured and found to be 55,375 and 42,639 mg/L, respectively. The sludge was kept in a plastic container and degassed by letting it self-digest the indigenous substrate, i.e., undigested manure, until biogas production was less than 1.0 mL/gVS according to Thaemngoen et. al. [27] This was done to minimize the

interference of end-product generation to that of the test substrate in the batch assays. TS and VS were then measured again to determine the amount for use as inoculant. Specific methanogenic activity of inoculum was 19.5 $mLCH_4/gCOD_{add}$.

2.3. Experiment 1: Batch study of pre-hydrolysis on effects of ORP and temperature on fermentative reactions

2.3.1. Pre-hydrolysis reactor configuration

Pre-hydrolysis (pHY) reactor was made of cylindrical glass with a working volume of 5.0 L (total volume of 6.25 L). The top of pHY reactor was closed with a solid PVC lid which housed four ports; (1) sampling port, (2) thermocouple, (3) oxidation-reduction potential (ORP) electrode, and (4) aeration port. The aeration port was connected to a diffuser submerged in the reactor. Air pump (TWIN, MEGA 6000, China) was controlled by the ORP controller (Cole-Parmer, 350 pH/ORP/Temperature 1/8-DIN, USA) sensed by the ORP electrode installed at mid depth in the pHY reactor. The reactor was placed on a magnetic stirrer (IKA®, C-MAGHS 7, Germany) for homogeneous mixing during experiment. Temperature in pHY reactor was controlled by a thermocouple and a temperature controller (Panasonic, AKT4R112200, Japan) whereas a heat source was provided by heat tape (AKO, 52346, Spain) wrapping around the reactor.

2.3.2. Experimental design of batch pre-hydrolysis

This study used only FMW (TS 1%) prepared according to Section 2.1 as substrate without any seeding in order to simulate the pre-hydrolysis tank on the field that receives only fresh manure wastewater (already containing natural live digestive microbes). Two experiments were carried out. The first was to determine the effects of ORP at four levels, i.e., natural anaerobic (no micro-aeration), −420, −360 and −300 mV. All were conducted at laboratory room temperature 28–30 °C. ORP was controlled by air injection through a micropore membrane diffuser. The original ORP of FMW was approximately 49 ± 10 mV and started to decrease from the beginning to reach the designed control values of −300, −360 and −420 mV at 9, 10 and 14 h, respectively. From the beginning, samples were collected every 6 h until reaching those target ORP values, where another sample was collected as a starting point of each treatment. Then, aeration by an air pump was automatically turned on by an electrical relay switch circuit whenever the reactor ORP was 10 mV below the target value. The air pump continued until the ORP reached 10 mV above the control value, then automatically turned off. During this period, the samples were collected at six-hour intervals until approximately 72 h in total run time. There were around ten sampling points in each run.

The second experiment was to determine the effects of temperature, which composed of four treatments at 30, 35, 45 and 55 °C. All were run at a fixed ORP of −420 ± 10 mV, which was identified as the best condition from the preceding experiment. Each pHY reactor was heated to its respective temperature within 10 min while ORP was continuously monitored. Similarly, the samples were taken every 6 h from the beginning and at the point when ORP reached −420 mV (at hour 14), then at an interval of 6 h afterwards. There were ten sampling time points in total at 0, 6, 12, 14, 20, 26, 38, 50, 62 and 74 h.

2.4. Experiment 2: Continuous digestion studies on effects of phase separation, pre-hydrolysis, digestion temperature, and sludge re-hydrolysis

2.4.1. System configuration

The reactor systems operated in this study were in either single-stage or two-stage configuration. In the single-stage configuration, only an anaerobic plug flow (PF) reactor was run whereas a pHY reactor was added as a front-end unit in the two-stage configuration setup. There were two identical reactor systems operated in parallel according to the conditions stated in Table 1 as single-stage (PF1 and PF2) and two-stage (pHY1 + PF1 and pHY2 + PF2) (Fig. 1). The pre-hydrolysis (pHY) reactor has a configuration and accessories as earlier described in Section 2.3.1 but with a smaller effective volume of 1.5 L. PF reactor was made of transparent PVC, which had an internal diameter of 10.2 cm and a length of 85.0 cm with the total and effective volumes of 7.0 and 5.0 L, respectively. Temperature of PF reactor was maintained by a heat tape connected to a thermocouple and a temperature controller. Biogas produced was collected in a gasbag (Tedlar®, 232-05, UK) which was connected to the reactor with low permeability tube (Masterflex, Silicone Tubing (Platinum) L/S®25, USA). The PF reactor was installed at horizontally decline angle of 10 degrees to provide the sludge accumulation zone. Underneath this zone, there was a sludge draining port that enabled sludge recycling to pHY reactor in Phase V of this experiment (see Table 1).

2.4.2. Digester system operations

At startup, all reactors were inoculated with the anaerobic sludge prepared in Section 2.2 to achieve an initial sludge concentration of 22,000 mg/L, then added DMW (1%TS) to the reactor at 25% of its volume per day. These PF reactors were operated at a hydraulic retention time (HRT) of 4 days corresponding to an organic loading rate (OLR) of 3.63 $kgCOD/m^3.d$ until reaching stable condition. This established a baseline piggery wastewater digestion performance of the original PF reactor (without retrofit). Later, pHY reactor operated at HRT 1 day was connected to each of the two PF reactors as two-stage configuration. DMW was then fed to the pHY reactors instead. The pHY effluents were sampled for analysis, and fed to their respective PF reactor. The operating schemes commenced in this experiment are divided into five phases as summarized in Table 1. In Phase I, both systems were operated as single-stage whereby only PF reactors were run under the identical condition. In Phase II-V, the two-stage configuration systems, designated pHY1 + PF1 and pHY2 + PF2, were run at different schemes with a fixed HRT and organic loading rate. By these

Table 1
Operating schemes of continuous anaerobic digester systems.

Experimental scheme	Operating conditions					System designation
	1st stage			2nd stage		
	Pre-hydrolysis reactors (pHY1 and pHY2)			Plug flow 1 (PF1)	Plug flow 2 (PF2)	
	ORP (mV)	Temp (°C)	Sludge recycle (%V/V_{PF})	Temp (°C)	Temp (°C)	
Phase I	n.a.	n.a.	n.a.	30	30	PF^{I}_{30C}
Phase II	Anaerobic	30	0	30	30	$pHY^{Anaerpnoc}_{30C} + PF^{II}_{30C}$
Phase III	-420 ± 10	30	0	30	30	$pHY^{-420mV}_{30C} + PF^{III}_{30C}$
Phase IV	-420 ± 10	45	0	35	45	$pHY^{-420mV}_{45C} + PF^{IV}_{35C}$ and $pHY^{-420mV}_{45C} + PF^{IV}_{45C}$
Phase V	-420 ± 10	45	10	35	45	${}_{35C}pHY^{-420mV}_{45C} + PF^{V}_{35C}$ and ${}_{45C}pHY^{-420mV}_{45C} + PF^{V}_{45C}$

Note: n.a. = "not available" because the systems were run as single-stage.
Phase II, pHY reactors were run without micro-aeration (no ORP control), which is designated as "Anaerobic". The ORP was −460 mV during such period.

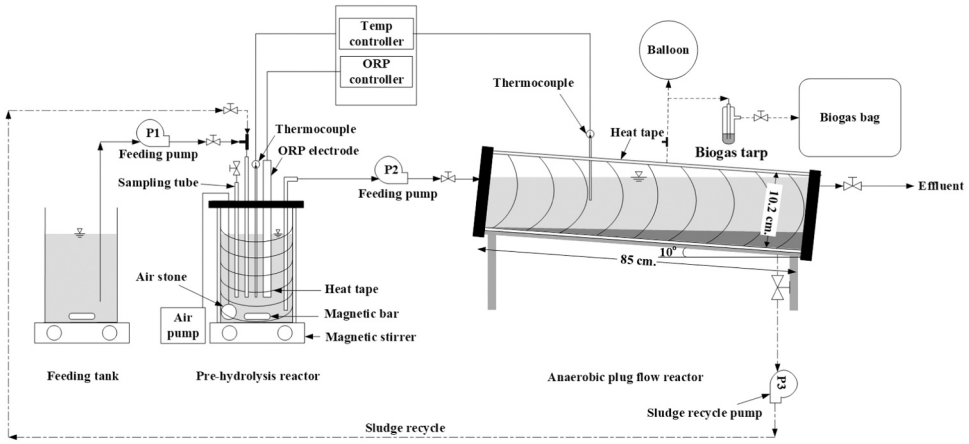


Fig. 1. Continuous two-stage anaerobic digestion system used in this study.

arrangements, both pHY reactors were run in identical conditions in Phase I-IV. Only in Phase V, they received the sludge returned from PF of different operating temperatures for sludge re-hydrolysis. Note that the pHY performance data reported herewith in Phase I-IV are from pHY1 reactor, and Phase V data are from pHY1 and pHY2 reactors separately. Data of pHY2 reactor in Phase I-IV were assessed constantly to ensure equivalent performance prior to changing the operating phase.

2.5. Analytical methods

Total solids (TS) and volatile solids (VS) were determined using Standard Methods for the Examination of Water and Wastewater [29]. Samples of swine manure were dried at 60 °C to constant weight and ground for analysis of elemental composition; carbon, hydrogen, nitrogen, oxygen and sulfur, by dynamic flash combustion method using CHNS-O Analyzer (CE Instruments Flash EA 1112 series, Themos Quest, Italy).

Liquid samples were measured for ORP and pH using the electrode from Cole-Parmer (Self-Cleaning Flat DJ/CPVC/Pt/10′/BNC, USA) and Mettler-Toledo (FiveEasy PlusTM, Germany), respectively. TCOD and SCOD were determined following the Standard Methods [29]. For analysis of individual volatile fatty acids (acetic, propionic, butyric and valeric acid) concentration, samples were filtered through a 1.2-μm GF/C for SCOD measurement, and through a 0.22-μm Teflon filters prior to injecting to the gas chromatography (GC7820A, Agilent technologies, USA) equipped with a flame ionization detector (FID) and a capillary column 30 m × 0.25 mm internal diameter with argon as a carrier gas. Alkalinity and total volatile fatty acids (TVFA) were analyzed by a direct titration method [30].

Biogas volume was measured daily using a high precision gas flow meter (Multi-chamber rotor, drum-type wet gas meter, Ritter, TG0/5, Germany). Biogas production was adjusted to STP (0 °C, 1 atm) in all experiments in this study. For methane content determination, the collected biogas was sampled and injected to the gas chromatography (GC7820A, Agilent technologies, USA) equipped with a thermal conductivity detector (TCD) and a stainless-steel packed column Hayesep Q80/100 (6 m × 1/8 in.) using helium as a carrier gas.

Microbial community analysis was carried out using Next Generation Sequencing (NGS) method. Sludge samples of 100 mL from Phase IV of the main reactors (PF reactor) during the continuous digestion study, operated at different temperatures (35 °C versus 45 °C), were taken at steady-state performance and kept in sterile plastic bottle at −20 °C until analysis. The samples were analyzed for bacterial (hydrolytic and acidogenic organisms) and archaeal (methanogenic organisms) communities and presented in terms of relative abundance. Genomic DNA was extracted from approximately 2 g of sludge using GF-1 Nucleic Acid Extraction Kits (Vivantis Technologies, Malaysia). PCR amplification was done twice. The first PCR was performed with a set of V3-V4 region primers (forward primer: 5′-TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGCCTACGGGNGGCWGCAG-3′, reverse primer: 5′-GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGGACT ACHVGGGTATCTAATCC-3′) and the PCR cycle condition started at 95 °C (3 min), then 25 cycles of 95 °C (30 s), 55 °C (30 s), 72 °C (30 s), and a final extension of 72 °C (5 min). The second PCR was performed with Nextera XT Index Primer (N7xx) and Nextera XT Index Primer (S5xx) (Illumina, USA) using the PCR cycle condition according to the Illumina's protocol. The purified amplicons were sequenced using Illumina MiSeq (reagent cartridge with 2 × 630-bp paired ends). The DNA were read using MiSeq reporter software and Greengenes database. The data obtained were demultiplexed and reads were then classified to different taxonomic levels.

2.6. Calculations

Hydrolysis and acidogenesis efficiencies were estimated based on the formulae adapted from Wu et al. [31]. Efficiency of the hydrolysis reaction is represented by a relative concentration of hydrolysis products (soluble organics) in terms of SCOD and CH_4 leaving the system to the total incoming organics according to Eq. (1). Acidogenesis efficiency is represented by a ratio of total volatile fatty acid concentration and CH_4 leaving the system lumped as COD to the total incoming organics, Eq. (2). These two terms were used to indicate the degree of the fermentation efficiency for pre-hydrolysis experiment (Section 2.3), in which there was no methane generation term.

$$\text{Hydrolysis Efficiency } (\eta_H) = \frac{(\text{mass SCOD} + \text{mass COD}_{CH_4})_{\text{eff.}} - (\text{mass SCOD})_{\text{inf.}}}{(\text{mass TCOD})_{\text{inf.}}} \quad (1)$$

where massSCOD is SCOD weight (g), massTCOD is TCOD weight (g) massCOD_{CH4} is CH_4 weight calculated as COD (g), massCOD_{VFA} is the sum of individual volatile fatty acids analyzed by GC expressed as gram

$$\text{Acidogenesis Efficiency } (\eta_A) = \frac{(\text{mass COD}_{VFA} + \text{mass COD}_{CH_4})_{\text{eff.}} - (\text{mass COD}_{VFA})_{\text{inf.}}}{(\text{mass TCOD})_{\text{inf.}}} \quad (2)$$

of COD calculated from the theoretical COD equivalents (g), inf. is influent, and eff. is effluent. The COD equivalents of acetic, propionic, butyric and valeric acid are 1.066, 1.512, 1.816 and 2.036, respectively.

2.7. Statistical analysis

All batch experiments were conducted in triplicate where means and standard deviations calculated are presented and discussed. The results from continuous anaerobic digestion reported were calculated from data at stable conditions, at least last three data points of each phase. All means were compared to determine the effect of the independent parameters in the performed experiments at a confidence interval of 95% ($\alpha = 0.05$) using Duncan's new multiple range (DMRT) by SPSS software version 11.0.

3. Results and discussion

3.1. Experiment 1.1: Effects of ORP on fermentative reactions

Hydrolytic bacteria are mostly aerobes, or facultative bacteria able to respire [15,32] with no or minimal loss of carbon through complete aerobic oxidation. SCOD release was thus used as an indicator of hydrolysis. Operation at ORP −300 mV required 6 h more to reach its maximum SCOD release while ORP −360 and −420 mV needed another 12 h to arrive at their maximums which are higher than that at −300 mV (Fig. 2a). The maximal SCOD release equivalently occurred at ORP −360 and −420 mV during hours 22 and 26 with the values of 2293 ± 92 and 2320 ± 80 mg/L, corresponding to hydrolysis efficiency (η_H) 0.026 and 0.027, respectively (Fig. 2b). This is 149.5% higher than the natural anaerobic (control) condition.

In the second step of AD, acidogenesis, the hydrolysates, e.g., monosaccharides, amino acid and SCOD, were converted to VFAs. From Fig. 2c and d, time to reach the peak TVFA concentration and peak η_A synchronized well with that of SCOD release and η_H in each of the ORP levels. This was due to that both reactions can be carried out by the same set of facultative bacteria [33]. The peak TVFA and η_A achieved were 12.5% and 59.2% higher than those in natural anaerobic condition, respectively. Even the worst η_H and η_A under micro-aeration (i.e. −300 mV) were still 63.2% and 20.4% better than the anaerobic (without aeration). Distribution of individual VFA at peak (Fig. 2d) did not vary by the ORP level applied. However, VFA data (Fig. S1 in Supplementary Material) beyond peak time points clearly showed differential disappearances across the treatments, suggesting that O_2 supplied may have oxidative preference among the individual VFAs. Larger VFA

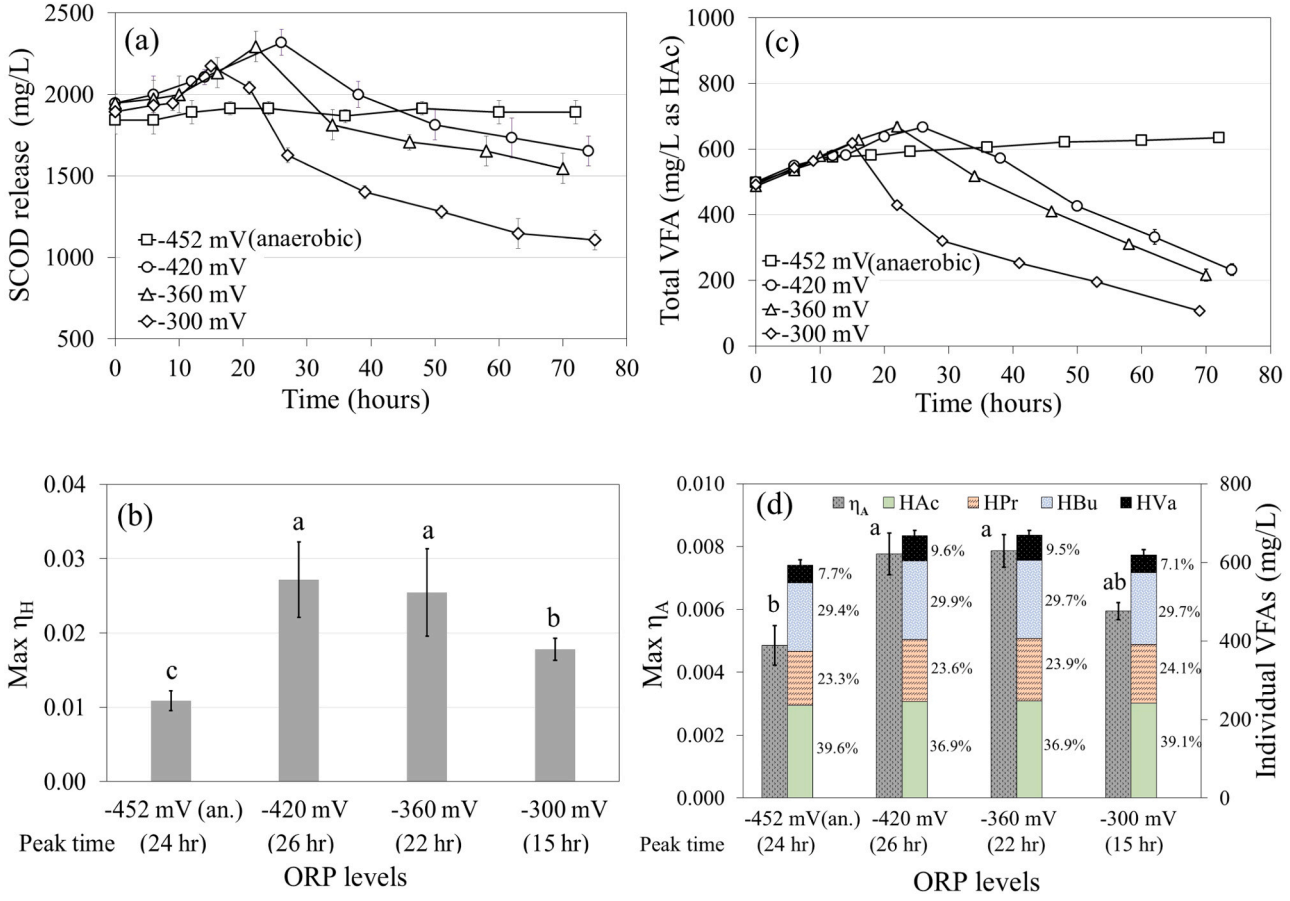


Fig. 2. Effects of ORP on key performance parameters of batch pre-hydrolysis reactor digesting piggery wastewater; (a) SCOD release vs. reaction time, (b) maximal hydrolysis efficiency (η_H) at different ORP levels, (c) total VFA concentration vs. reaction time, and (d) maximal acidogenesis efficiency (η_A) with individual VFAs profile at different ORP levels. Different letters indicate significant difference of the means at $\alpha = 0.05$.

(valeric and butyric acids) were unproportionally targeted for oxidation. This could be beneficial to strict anaerobic culture such as methane producing digester since the larger VFA are more inhibitory even at low concentration [34].

These results demonstrate the effectiveness of the ORP control strategy by micro-aeration to hydrolysis and acidogenesis in relation to reaction time. It clearly differs from the direct micro-aeration in the methanogenic reactor [11] that may cause undesirable impacts to the methanogenic reaction by inhibiting methanogens with the oxygen supply was well as diluting methane in the biogas. In this present study, all statistical comparisons confirmed the significant improvements ($\alpha = 0.05$) in performance of ORP controlled hydrolysis reactors under batch operation. Instead of the equivalent air or oxygen to COD ratio such as reported in Barati rashvanlou et al. [13] and Fu et al. [35], our method disregards such ratios but controls on the basis of redox state. The ORP is directly attributed to the biochemical condition that the facultative microorganisms' activities and their extracellular enzyme production, e.g., amylase, cellulase, and protease are driven [11]. However, aeration beyond the optimal time led to the continual decline in SCOD availability signifying the loss of carbon through heightened respiration and complete oxidation to CO_2 [33], from which it must be avoided. Implication of this data set lies in the application of micro-aeration at an existing wastewater collection tank (a tank with only level control) of an anaerobic treatment system in animal farms. Results in the next section on the effects of temperature on pre-hydrolysis will further our understandings on the conditional control of this wastewater collection tank and possible gain from its retrofit.

3.2. Experiment 1.2: Effects of temperature on fermentative reactions

Impacts of temperature on hydrolysis and acidogenesis reactions in manure wastewater were evaluated at mesophilic (30 and 35 °C), sub-thermophilic (45 °C) and thermophilic (55 °C) conditions under a fixed ORP at −420 mV (the best condition determined from the preceding experiment). It was observed that the time to reach ORP −420 mV was shortened at most by only 40 min, insignificant to the total reaction time. At 6 h of sampling interval after reaching ORP −420 mV, the SCOD peak time in this experiment fell on the same time point at 26 h from start (Fig. 3a).

Results in Fig. 3a clearly show that SCOD release increased with higher temperatures. The maximal η_H in this study was 0.051, found at 55 °C and 26 h from start (Fig. 3b), This η_H was higher than at 30 °C by as much as 88.9%. The underlining reason for heightened hydrolytic activity at thermophilic temperature is the acceleration of temperature-dependent enzymatic reactions acting on the surface of the swine manure particles [36], together with the microbial cells activity that induce higher release of extracellular hydrolytic enzymes [19]. Moreover, the partial modification of substrate structure, particularly the particulate portion, even by low-range hydrothermal treatment also made it more susceptible to enzymatic attacks at a more porous physical matrix [28]. Above those of the mesophilic temperature, η_H at sub-thermophilic and thermophilic temperatures yet appeared to be statistically similar ($\alpha = 0.05$). A raise of temperature from 45 to 55 °C was not able to provide a significant benefit in piggery wastewater hydrolysis beside the certainty of larger investment on insulation and energy to operate a thermophilic system. This phenomenon was perhaps related to the nature of this piggery waste stream that is quite diluted and less digestible as it was partially digested in the animal intestinal

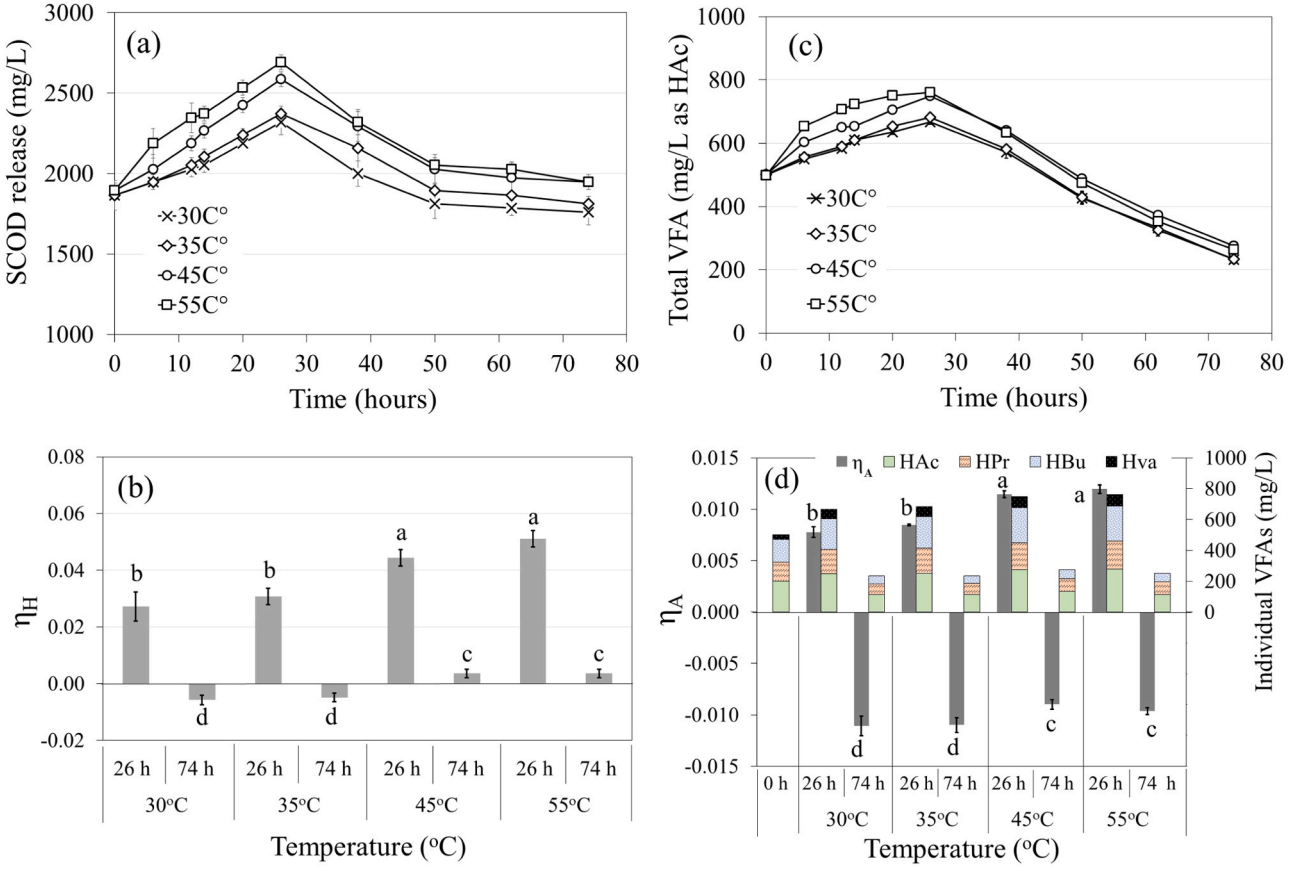


Fig. 3. Effects to temperature in batch pre-hydrolysis reactor on (a) SCOD release, (b) hydrolysis efficiency (η_H), (c) total VFA produced, and (d) acidogenesis efficiency (η_A) with individual VFAs profile at the start (0 h), peak η_H time point (26 h) and end of batch (74 h). Different letters indicate significant difference of the means at $\alpha = 0.05$.

tract. The highly digestible materials such as food waste and soft biomass were more susceptible to temperature [24,37]. The gap in fermentative performance under sub-optimal temperature ranges of various substrates still require more investigations that it may defy our standing belief of their inefficacy. Regarding the after peak event, i.e. after 26 h, η_H exhibited a severe down trend to the end of batch (74 h). This constituted the losses of SCOD at 21–29%.

The impact of temperature on acidogenesis reaction is illustrated in Fig. 3c and d. Similar to hydrolysis, TVFA and η_A increased with the elevated temperatures. The rate of TVFA generation represented by a line slope in Fig. 3c accelerated with temperature. η_A data confirmed that sub-thermophilic temperature was able to give an advantage to the pre-hydrolysis reactor over the mesophilic one. Distribution of individual VFAs had not played an important role in this regard (Fig. 3d). On the way to peaks, pH gradually dropped as a result of enhanced TVFA generation by the degradation of largely carbohydrates until later when protein started to degrade more, giving off ammonia alkalinity, and pH started to stabilize. TVFA to alkalinity ratio prior to η_A peak point was in the vicinity of 0.75–0.80 before dropping off to around 0.2 at the end of batch (72 h). Nevertheless, the reactor would not turn into methanogenic reactor as methanogens are still inhibited by ORP control regime and no accumulation of anaerobic culture sludge in the reactor. In this circumstance, the hydrolytic and acidogenic bacteria coming with the fresh manure are still active by their ability to tolerate oxidative-reductive environmental fluctuations and pH range of 4–7 [25].

Based on SCOD release data time lapse in Fig. 3a, pHY at 55 °C took only 23 h to reach the peak of pHY 45 °C at 26 h (2587 mg/L), that is approximately 11% faster. This number could infer to a potential reduced volume requirement for pre-hydrolysis tank at 55 °C while still obtaining an equivalent performance to the 45 °C based on η_H. Nevertheless, such deficiency is not statistically significant and may be compensated by various digester operational schemes, which will be critically assessed later in this paper.

For field application, our experience dictates that the waste heat from biogas engine (a combined heat and power generation unit) on a pig farm in Thailand only provides enough heat to raise temperature of a wastewater collection tank, located in front of a plug flow digester, to the sub-thermophilic temperature, around 45 ± 3 °C depending on flow variation. This tank generally acts as a buffer sump. With a volume to bear a hydraulic retention time up to 24 h, it presents a great potential to retrofit to a pre-hydrolysis tank. Results in the ensuing experiments will elucidate such feasibility in four schemes under continuous operations.

3.3. Experiment 2: Continuous anaerobic digestion at various operational schemes

In this experiment, effects of pre-hydrolysis tank installation, ORP controlled micro-aerations, sub-thermophilic temperature, and sludge re-hydrolysis could be quantified by comparing Phase I vs. II, Phase II vs. III, Phase III vs. IV, and Phase IV vs. V, in order. Sub-thermophilic temperature of 45 °C was chosen according to the reasons previously mentioned.

3.3.1. Performance of pre-hydrolysis reactor (1st stage)

3.3.1.1. Acid accumulation. In Phase II, when pHY reactor operated at natural anaerobic condition (no micro-aeration; ORP −460 mV) and 30 °C (designated as $pHY_{30C}^{Anaerobic}$) was placed in front of the PF reactor (PF_{30C}^{II}), pH dropped to 7.06 ± 0.05 while TVFA rose to 238 ± 14 mg/L as HAc at the end of Phase II (Fig. 4a). When micro-aeration was implemented (Phase III) at the ORP from Experiment 1.1 of −420 mV, $pHY_{30C}^{-420\ mV}$ was able to increase TVFA by 7.1% to 255 ± 12 mg/L as HAc and pH further dropped to 6.78 ± 0.03. It is worth noticing that TVFA/ALK in pHY reactor of around 0.6 ± 0.1 posed no threat to the subsequent PF reactor due to its larger volume (HRT 4 days). Ammonia (slightly increased) and alkalinity from the degradation of protein remaining in the pig manure [38] also played a part in maintaining buffer capacity of the reactor. The small losses of ammonia and dissolved matters, i.e., organic acids and soluble COD were due to the drying process when the fresh manure was preserved.

In Phase IV (Fig. 4a), temperature of pHY reactors was raised to 45 °C. TVFA in $pHY_{45C}^{-420\ mV}$ increased only slightly by 5.9% to 270 ± 12 mg/L as HAc while pH drop was insignificant compared to 35 °C (Phase III). The largest impact yet to acidogenesis efficiency, signified by a drastic change in TVFA generation, was observed in the sludge re-hydrolysis scheme when the sludge from PF reactor bottom was returned to pHY reactors in Phase V. The sludge from $PF_{45\ C}^{V}$ became more digestible and caused a surge in TVFA by 32.2% to 357 ± 14 mg/L as HAc that also brought pH down further to 6.53 ± 0.02, compared to TVFA 305 ± 16 mg/L as HAc and pH 6.63 ± 0.03 with the sludge return from PF_{35C}^{V}. Acidogenesis efficiencies of pHY reactor in Phase II, III, IV, and V were 0.0012, 0.0020, 0.0030, and 0.0033 ($_{35C}pHY_{45C}^{-420mV}$) and 0.0048 ($_{45C}pHY_{45C}^{-420mV}$), in order. Micro-aeration in combination with sub-thermophilic temperature could improve η_A in pHY reactor by a total of 150%. And with sludge re-hydrolysis, significant improvements of 175% and 300% can further be gained owing to the extended hydrolysis treatment from both stages prior to re-enter the pHY reactor.

3.3.1.2. Hydrolysis efficiency. From Fig. 4b, SCOD of the feed and effluent of $pHY_{30C}^{Anaerobic}$ were quite consistent at 639–653 mg/L (Phase I-II). In terms of hydrolysis, there was no significant difference by installing pHY reactor without any control. This was also confirmed by their similar methane yield data which will be discussed in Section 3.3.2. In Phase III, ORP controlled micro-aeration ($pHY_{30C}^{-420\ mV}$) had clearly displayed a clear advantage in hydrolysis efficiency. The phenomenon appeared to conform and even surpass that in the preceding batch experiment (+149%). η_H in continuous pHY reactor improved as much as 233% corresponding to the major incremental SCOD release from 653 ± 46 to 880 ± 40 mg/L. The on-demand supply of air in response to the replenishment of the substrate in this semi continuous feeding mode not only was proven to enhance hydrolytic activity [33] but also have provided successive biostimulation that caused proliferation of the selected groups of the fermentative microbial culture over time in the retained biomass [35]. Molecular oxygen supplied in air is essential for facultative organisms to switch to the more energetic aerobic pathways [39] with higher cell activity and enzyme production [35]. Nevertheless, over supply of oxygen by either excessive amount or extended period will prompt cells to biochemically respire with the expense of organic carbon in their environment, thus diminishing the cause of pre-hydrolysis.

Under sub-thermophilic temperature, hydrolysis had improved. $pHY_{45C}^{-420\ mV}$ was able to raise SCOD release by 10.8% over $pHY_{30C}^{-420\ mV}$ (Fig. 4b). Regarding time for microbial acclimation, it took only 20 days to reach stable performance after imposing the new temperature. This quick adaptation was a good trait of facultative fermentative bacteria to temperature, but it was even shorter with the exposure to air from Phase II to Phase III. This suggests an un-interrupting implementation to the existing system with neglectable lag time. Unlike the pHY reactor, the methanogenic reactors are more sensitive to environmental changes and may take much longer time to adapt. Switching and adapting to the sub-thermophilic temperature of the plug flow reactors in this study had revealed some interesting information that ultimately linked to its biomethanation efficiency as it would brought into discussion in the next section.

In Phase V, effluent of pHY receiving sludge returned from PF_{45C}^{V} had higher SCOD concentration compared to that from PF_{35C}^{V} (Fig. 4b). Re-hydrolysis of the sludge from PF_{45C}^{V} is proven more effective although the sludge biodegradability ratio (VS/TS) was approximately 3–5% lower than that of PF_{35C}^{V} due to the higher hydrolysis in PF_{45C}^{V}. This flow

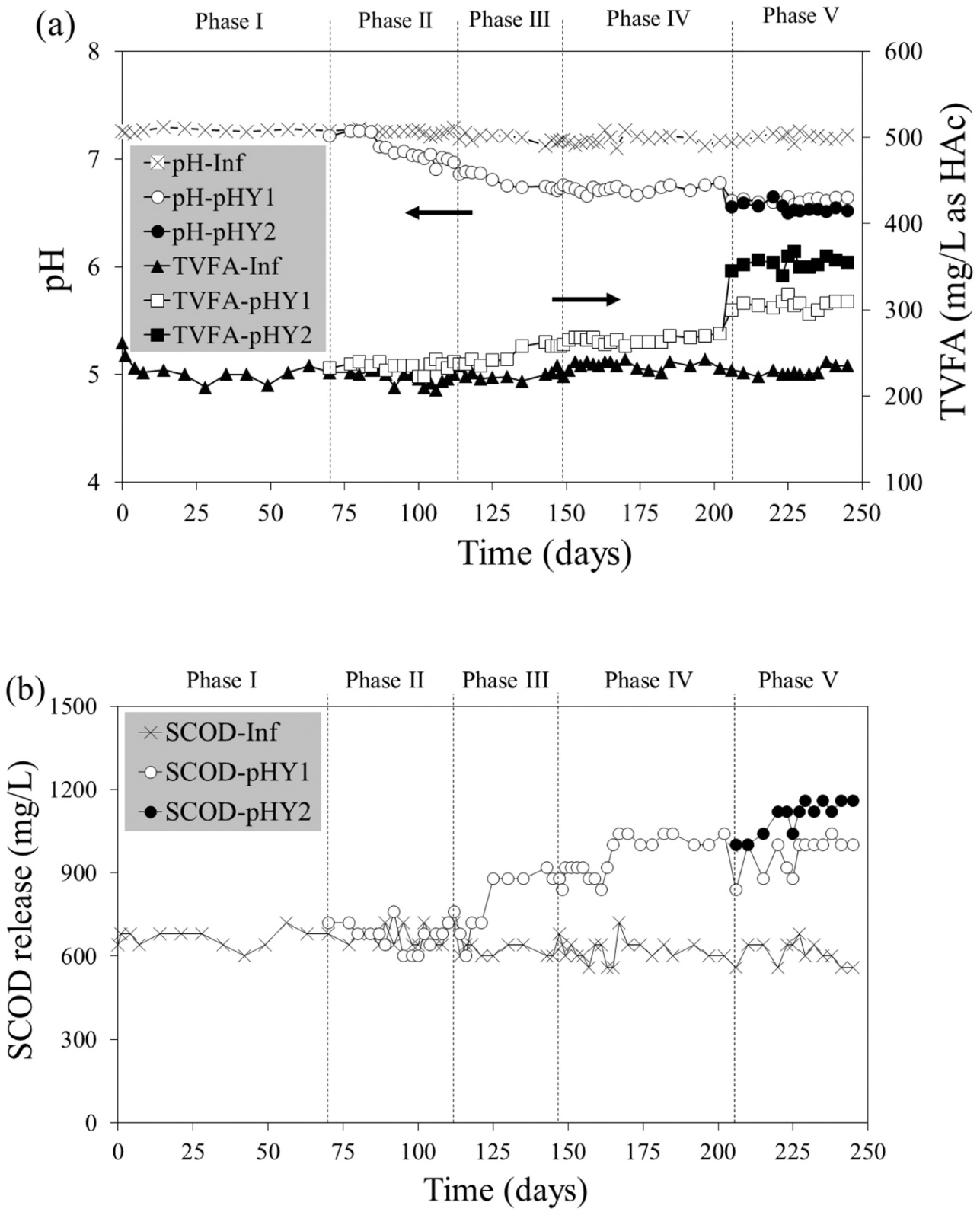


Fig. 4. Performance of pre-hydrolysis (pHY) reactor in continous operations; (a) pH and TVFA variation, and (b) SCOD release. (Note: there is no pHY performance data in Phase I since the systems were run as single-stage configuration, and there are two separate pHY performance lines in Phase V as they received different returned sludges from two PF reactors of different operating temperature. See Table 1 for details.).

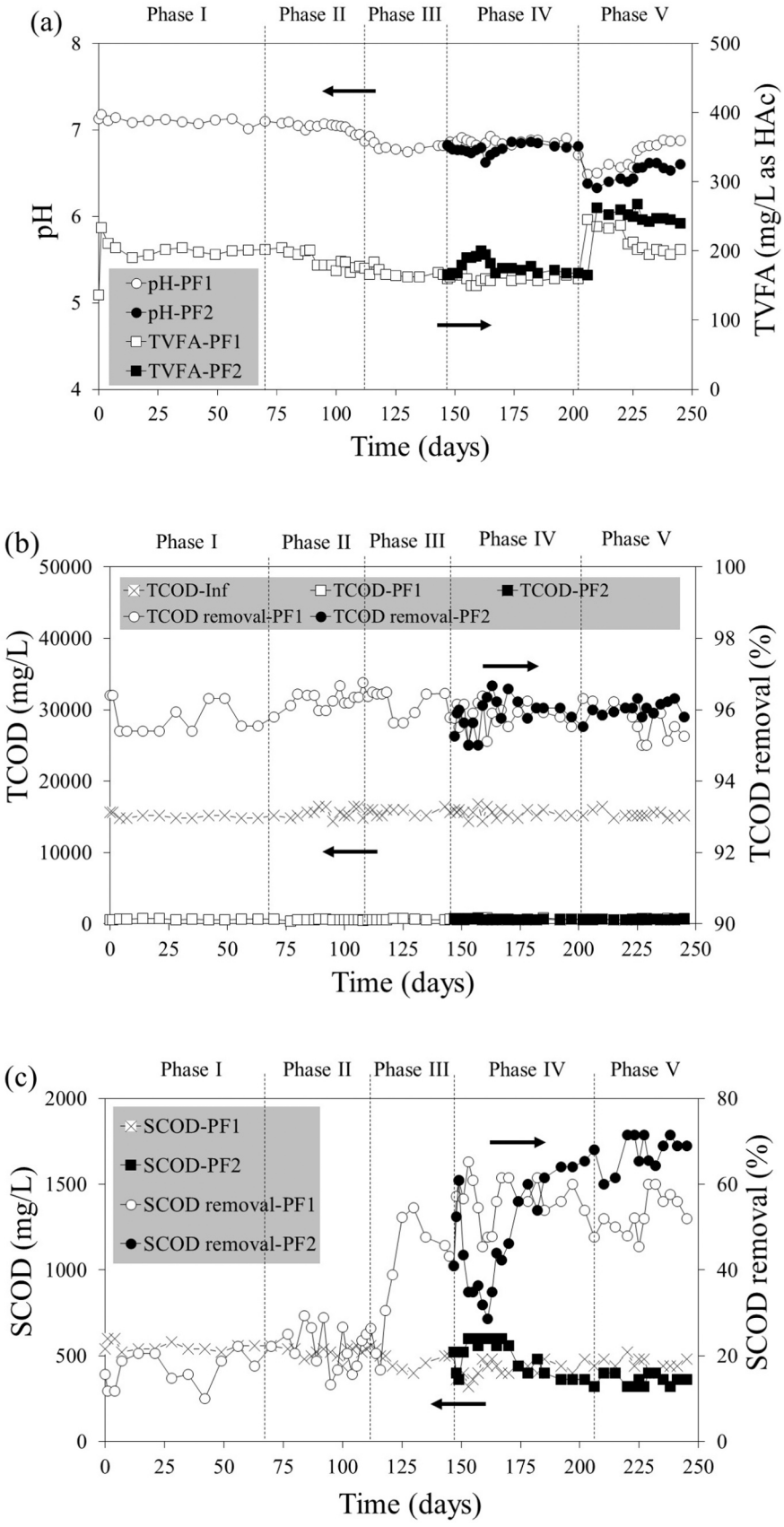


Fig. 5. Performance of plug flow (PF) digester in continous operations; (a) pH and TVFA of effluent, (b) total COD concentration and removal, and (c) soluble COD concentration and removal. (Note: the systems were operated in single-stage configuration in Phase I and two-stage configuration in Phase II-V. The two systems were operated identically in Phase I-III and differently in Phase IV and V. See Table 1 for details.).

from PF^{V}_{45C} did not need time to heat up and it should theoretically be more susceptible to hydrolysis by its exposure to the higher temperature, which is in agreement with the TVFA data in Fig. 4a. HRT at 4 days and 45 °C in PF reactor played a significant role in overall digestion in this case. The benefit truly lies in the return of the degradable organics for re-digestion through the whole system. Since the system is a plug flow type, the logical way of mixing in order to utilize the overall digester volume is to recycle the concentrated degradable portion of remaining substrate by the longest stretch of the system. In this study, the ratio of sludge return flow was set at 10%, although smaller recycle flow may be more desirable since lower energy expenditure is required. Optimization of recycle flow ratio in this system should be further investigated in relation to operating temperature.

3.3.2. Performance of anaerobic plug flow digester (2nd stage)

3.3.2.1. System stability. Fig. 5 shows that merely placing pHY reactor to the existing PF reactor did not clearly cause changes in buffer balance in the second stage PF reactors (Fig. 5a). A slight drop in pH was observed as TVFA from the system declined from 200 ± 17 in Phase I to 180 ± 15 mg/L as HAc in Phase II. By controlling ORP, pH in $PF^{III}_{30\ C}$ began to drop from around neutral to 6.80. TVFA in effluent, however, continued to decline in Phase III as the ORP-controlled pHY helped hydrolyze the substrate making it easier for PF reactor to process.

Immediately after entering Phase IV (Fig. 5a), the transient period started at day 147 where the residual TVFA in the effluent from PF^{IV}_{45C} apparently jumped compared to PF^{IV}_{35C} which was accompanied with a slight drop in pH. This temperature shock to methanogens had temporarily lowered their activities to convert fatty acids to methane but it was, however, overcome within only 27 days in this operating scheme. Similarly, the sludge recycle in Phase V appeared to cause a significant TFVA accumulation from 168 ± 15 and 166 ± 13–245 ± 13 and 200 ± 14 mg/L as HAc for PF^{V}_{45C} and PF^{V}_{35C}, respectively, at steady state. pH drop occurred and eventually stabilized at 6.59 ± 0.04 in PF^{V}_{45C} and 6.85 ± 0.02 in PF^{V}_{35C}, which were still within the suitable range for methanogenic reactions at 6.5–7.5. The buffer capacity (VFA/ALK) at this phase was still in check around 0.20, still lower than the recommended limit of 0.4 for stable methanogenic reactor [40]. In another aspect, the impact of this sludge recycling is somewhat comparable to the increase of organic loading to the system. Only in this scheme, it utilized the internal under-digested substrate as carbon source.

3.3.2.2. COD removal. Fig. 5b shows that the overall TCOD removals were in a range of 95–97% across five different phases in the experiment. Since the manure wastewater was prepared from dry manure powder which largely did not dissolve in water, the particulates tended to settle in PF reactors. TCOD removal was, thus, predominantly carried out by physical sedimentation and could not fully reflect the biological digestion. By converting a single-stage PF reactor to two-stage pHY + PF, some enhanced physical TCOD removal (by sedimentation) was observed, as depicted in Fig. 5b. Other than that, the effects of ORP, temperature and sludge recycle on TCOD were not clear. Thus, SCOD data were brought into analysis (Fig. 5c). SCOD of the feed around 640 mg/L was reduced only 22.4% in Phase I and II. When ORP-controlled micro-aeration was in effect, SCOD of PF influent was greatly increased (Fig. 4b) while still generating effluent with consistent SCOD, causing a jump in SCOD removal to 47.8%. Methanogenic shock by the elevated temperature in Phase IV brought the PF reactors to a transient state with unstable organic removal performance. At the end of Phase IV, SCOD removal of PF^{IV}_{45C}, however, reached 64.6% while PF^{IV}_{35C} obtained only at around 55.9%. Although recycling of sludge to pHY reactors did not show a clear sign of improved contaminant removals in both TCOD and SCOD terms, the advantage with this higher sludge degradation can actually materialize when a much lower sludge disposal is required over time. But in terms of energy recovery, the direct benefit can be measured through the observed methanation performance described in the following section.

3.3.2.3. Methane yield. Methane yield is a term defined by the volume of methane per unit mass of COD removed. Comparisons between the phases at $\alpha = 0.05$ were performed to identify the effectiveness of each scheme (Fig. 6a). It is worth noticing that practical methane yields obtained from this experiment are in the range of 120.3–144.6 $mLCH_4$/gCOD which is lower than hypothetical yield (BMP) and theoretical yield of 167–177 and 350 $mLCH_4$/gCOD, respectively. This is due to the fact that particulate COD is dominant in this wastewater. This type of COD can only be partially degraded, in contrast to the soluble substrate that can be more readily converted to methane. This gap of observed and theoretical methane yields has been shown significant in various high solid wastewaters [41]. By comparing Phase I vs. II, merely installing an uncontrolled pre-hydrolysis tank to PF reactor statistically did not give any methanogenic advantage; 120.3 ± 5.1 vs. 120.8 ± 3.5 $mLCH_4$/gCOD in Phase I and II, respectively. Although TVFA concentration entering PF reactor was higher compared to the raw manure wastewater (Fig. 6a), sufficient digestion time in PF reactor without pHY reactor could still compensate such benefit under this relatively low organic loading condition, especially when buffer capacity was still in excess [42]. Beyond that phase, each strategy applied yielded statistically significant improvements. While ORP-controlled micro-aeration alone (Phase III) could increase methane yield from 120.8 to 123.6 $mLCH_4$/gCOD (+2.3%), the additional temperature increase to mesophilic and sub-thermophilic (Phase IV) had added another 4.1% and 10.0%, respectively. Finally, the returned sludge re-hydrolysis (Phase V) pushed this methanogenic efficiency by another 4.9% and 6.3% for PF^{V}_{35C} and PF^{V}_{45C}, respectively. Overall, the compound improvement is 20.4% from the original one-stage digestion system based on methane yield (120.1 vs. 144.6 $mLCH_4$/gCOD). It is noted that this cumulative advantage in biomethanation efficiency is achievable without the help of external heat source.

3.4. Cost-benefit analysis

Preliminary cost-benefit evaluation is performed based on the results from the study and construction and operating costs at the studied pig farm in Ratchaburi province during the study time period (Year 2019–2020). A pig farm at Ratchaburi province generated wastewater at 450 m^3/d on average with TS 1%. By the experimental results, additional methane yield from retrofit of + 20.4% (0.1203 vs 0.1446 L/gCOD) would equate to 159.6 m^3 CH_4/d or 47,893 m^3 CH_4/yr (300 operating days per year). At approximately 60% CH_4 in biogas produced and biogas-to-electricity engine efficiency 1.9 kWh/m^3_{biogas}, the approximate energy gain from enhanced methane production would be 545,979 MJ/yr or 151,661 kWh/yr. With the electricity cost at 4.2 Baht/kWh, the potential annual benefit from this improvement would be 507,595 Baht/yr (32 Baht/USD). The approximate costs of retrofit by adding a pre-hydrolysis tank and exhaust gas and cooling water heat recovery system with pumps and piping was 1,970,000 Baht. Assuming the operating cost of two hot water pumps (8.5 hp or 6.3 kW) (0.9 × 6.3 kw × 300 d × 18 h/d × 4.2 Bath/kWh) = 129,380 Baht/yr. Thus, the simple payback period is approximately 3.88 years.

3.5. Changes in microbial communities at sub-thermophilic temperature

Employing the Next Generation Sequencing method, microbial community in the system was measured from the sludge samples collected from PF^{IV}_{35C} and PF^{IV}_{45C} at their stable operations toward the end of Phase IV. The original inoculum in this study was from anaerobic digester operated at ambient temperature (28–32 °C), thus, the microbial population would closely resemble those operated at 35 °C in the

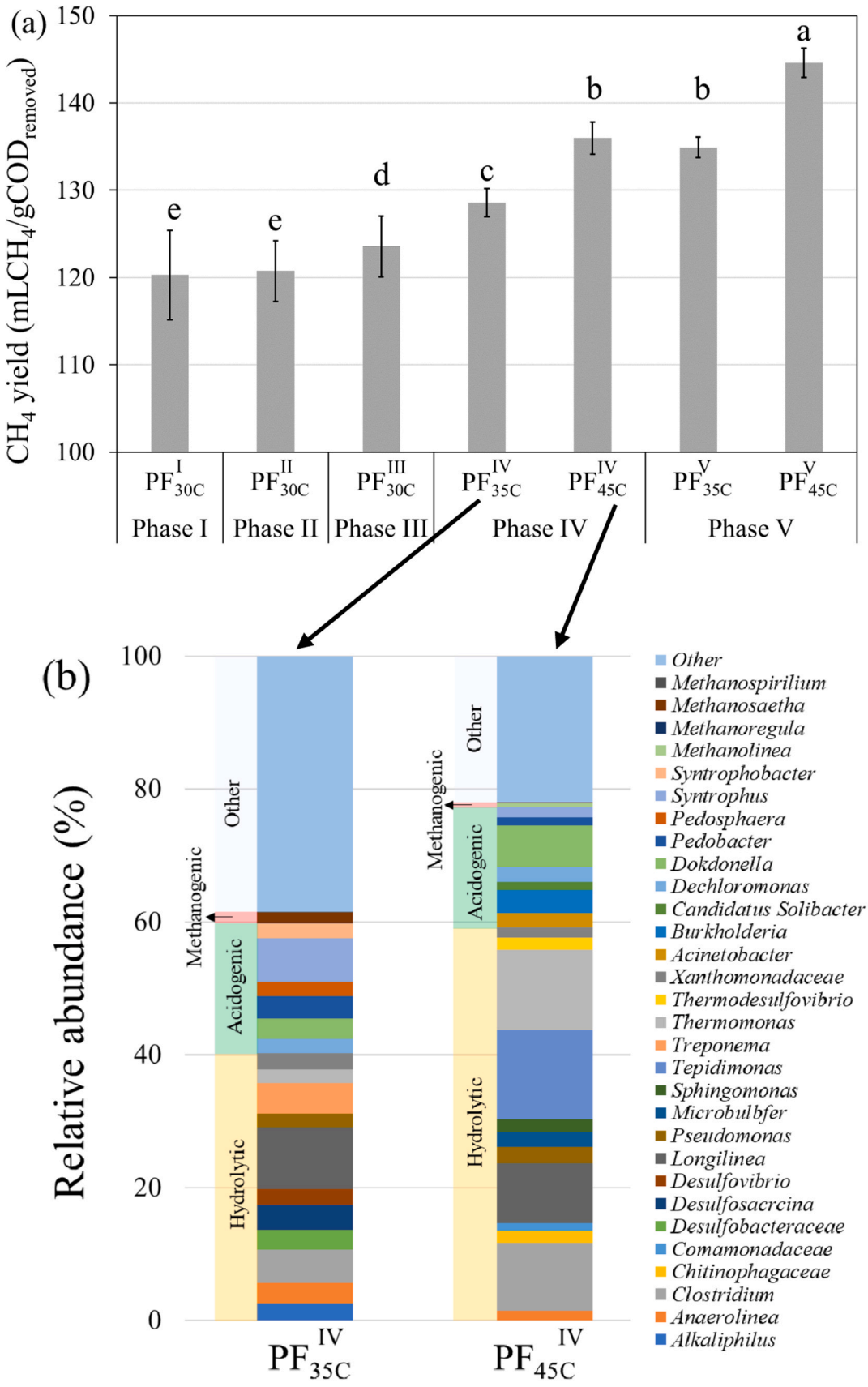


Fig. 6. Biomethanation efficiency of piggery wastewater at various operating schemes represented by (a) average methane yield, and (b) microbial community composition in plug flow digesters operated at 35 °C (PF^{IV}_{35C}) and 45 °C (PF^{IV}_{45C}) at the end of Phase IV. Different letters indicate significant difference of the means at $\alpha = 0.05$.

experiment.

From Fig. 6b, PF^{IV}_{35C} contained hydrolytic and acidogenic bacteria around 40.2% and 17.5%, respectively. The distinct hydrolytic bacteria in PF^{IV}_{35C} are such as *Longilinea* spp. (8.7%) and sulfate reducing bacteria *Desulfobacteracea* family (9.1%) of *Desulfosacrcina* spp. and *Desulfovibrio* spp. These sulfate reducing bacteria are good hydrolyzers for cellulose [43] with an optimum growing temperature in mesophilic range [44]. Their presence was replaced with the thermophilic genus of *Thermodesulfovibrio* spp. in the sub-thermophilic reactor PF^{IV}_{45C}. Data show the shift of a major microbial proportion of hydrolytic bacteria to 59.2% in PF^{IV}_{45C}.

It was clear that the hydrolytic group has expanded when the temperature rose. The hydrolytic bacteria *Clostridium* spp. and *Thermomonas* spp. had spawned unproportionally by 2.1 and 6.0 times over those in PF^{IV}_{35C}. Even though *Clostridium* are predominant in cellulytic anaerobic environments and proven highly active in both mesophilic and thermophilic environments [25], results suggest that they can thrive better in the sub-thermophilic temperature. Moreover, there was a higher diversity of other hydrolytic groups in PF^{IV}_{45C} such as *Sphingomonas* spp. the lignocellulose degraders [45] and *Tepidomonas* spp. the starch hydrolyzer [46], as well as *Thermomonas* spp. and *Thermodesulfovibrio* spp. the thermophiles whose enzymes are thermostable and more active than mesophilic enzymes [19]. It was, however, found that PF^{IV}_{35C} had a larger relative abundance of methanogens than PF^{IV}_{45C} (Fig. 6b). The methanogenic diversity in PF^{IV}_{35C} (three genera; *Methanospirilium* ssp., *Methanosaeta* ssp. and *Methanoregula* ssp.) was also higher than PF^{IV}_{45C} (two genera; *Methanospirilium* ssp. and *Methanolinea* ssp.). This data coincides with the slightly higher methane content in the biogas produced from PF_{35C}, i.e., 60.7 $\pm$ 2.3% vs. 58.4 $\pm$ 2.7% during Phase IV. This finding is consistent with the other research where the methane content in thermophilic reactor was generally lower than mesophilic reactors despite the fact that the thermophilic methanogens are biochemically more active in processing their substrates [47], which ultimately gave higher overall methane production in this study. The preference of methanogens to mesophilic temperature range is known to bring about the more diverse groups and higher abundance in this environment [48].

4. Conclusion

Different strategies applied to piggery wastewater digestion showed notable improvements. Micro-aeration through ORP control was proven an effective means to regulate fermentative reactions. Wastewater pre-hydrolysis and sludge re-hydrolysis under sub-thermophilic temperature (45 °C) demonstrated promising results to enhance overall digester performance. In continuous digester operations, a combination of ORP control, sub-thermophilic temperature, and returned sludge achieved an increase in methane yield as high as 20.4%. Enlarging hydrolytic bacteria abundance in sub-thermophilic reactor played an important role in this improvement. The investment on the proposed retrofit was estimated to have a payback period less than four years. The selected scheme is under investigation at a pilot swine farm in Thailand.

CRediT authorship contribution statement

Worakan Chetawan: Investigation, Data curation, Formal analysis, Writing - original draft. **Kanyarat Saritpongteeraka:** Methodology, Data curation, Project administration. **Arkom Palamanit:** Investigation, Conceptualization. **Sumate Chaiprapat:** Conceptualization, Funding acquisition, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jece.2021.105620.

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